



Junjie Nian 年骏杰

B.Eng. in Artificial Intelligence, Fudan University · Exchange Student at UC San Diego (Fall 2026)

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PROFILES [Google Scholar](#) · [DBLP](#) · [Semantic Scholar](#) · [ACL Anthology](#) · [ORCID](#)

ABOUT

Undergraduate researcher in Artificial Intelligence at Fudan University and exchange student at UC San Diego (Fall 2026). At FudanNLP ALEX Lab, advised by Prof. Yixin Cao, I study reasoning processes, agent reliability, and mechanistic interpretability.

EDUCATION

2024.9 – Present **Fudan University** — B.Eng. in Artificial Intelligence
School of Computer Science.

Fall 2026 **UC San Diego** — Exchange Student
Computer Science & Engineering.

2021.9 – 2024.6 **The High School Affiliated to Fudan University**

RESEARCH EXPERIENCE

2025.10 – Present **FudanNLP ALEX Lab** — Undergraduate Researcher
Advised by Prof. Yixin Cao. Research on Large Reasoning Model capabilities, agent systems, and mechanistic interpretability.

2025.3 – 2025.6 **Embodied Intelligence Research Project** — Tengfei Innovation Program
Advised by Prof. Shang Li. Built small-scale robots with asynchronous servo motor control and embedded development board integration.

PUBLICATIONS

Kang Chen*, **Junjie Nian***, Yixin Cao, Yugang Jiang. “Disagree to Explore, Agree to Commit: Routing-Guided Test-Time Scaling for Software Agents.” **Preprint, 2026. [arXiv] [Code]**

Kang Chen, Minshen Yu, **Junjie Nian**, Yaoning Wang, Yixin Cao, Yugang Jiang. “Does the Same Token Mean the Same State? MoE Routing as Signal for Reasoning Control.” **Preprint, 2026. [arXiv] [Website]**

Junjie Nian*, Kang Chen*, Ge Zhang, Yixin Cao, Yugang Jiang. “TraceGraph: Shared Decision Landscapes for Diagnosing and Improving Agent Trajectories.” **Preprint, 2026. [arXiv] [Code] [Trajectory Lens]**

Kang Chen*, **Junjie Nian***, Yixin Cao, Yugang Jiang. “SliceGraph: Mapping Process Isomers in Multi-Run Chain-of-Thought Reasoning.” **Preprint, 2026. [arXiv] [Code]**

Kang Chen*, Zhuoka Feng*, Sihan Zhao, Kai Xiong, **Junjie Nian**, Yaoning Wang, Changyi Xiao, Yixin Cao. “NEX: Neuron Explore-Exploit Scoring for Label-Free Chain-of-Thought Selection and Model Ranking.” **Preprint, 2026. [arXiv]**

Kang Chen*, Fan Yu*, **Junjie Nian**, Sihan Zhao, Zhuoka Feng, Zijun Yao, Heng Wang, Minshen Yu, Yixin Cao. “Thinking Traps in Long Chain-of-Thought: A Measurable Study and Trap-Aware Adaptive Restart.” **ACL 2026 Findings. [arXiv] [ACL Anthology]**

Zhuoka Feng*, Kang Chen*, Sihan Zhao†, Kai Xiong†, Yaoning Wang†, Minshen Yu, **Junjie Nian**, Changyi Xiao, Yixin Cao, Yugang Jiang. “ARM: Role-Conditioned Neuron Transplantation for Training-Free Generalist LLM Agent Merging.” **EMNLP 2026 Oral. [arXiv] [Website]**

* Equal contribution † Equal second-author contribution

HONORS & AWARDS

Duckbill Special Scholarship, Fudan University **2025**

Third Prize, China Undergraduate Mathematical Contest in Modelling, Shanghai **2025**

Third Prize, The Chinese Mathematics Competitions for College Students, Shanghai **2025**

Freshman Scholarship, Fudan University **2025**

First Prize, Tengfei College Entrance Exam Freshman Scholarship, Fudan University **2024**

SKILLS

Programming: Python, C/C++, JavaScript

Machine Learning & Scientific Computing: PyTorch, NumPy, SciPy

Web Development: React, FastAPI, Three.js, HTML/CSS

Tools: Git, Docker, LaTeX

ENGLISH PROFICIENCY

TOEFL **111** CET-6 **619** CET-4 **636**